

Quantitative recurrence properties for piecewise expanding maps on $[0, 1]^d$

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Outline

- 1 Motivation
- 2 Results
- 3 Ideas and proofs

Poincaré's recurrence theorem

- (X, d) : separable and complete metric space.
- $T : X \rightarrow X$.
- μ : T -invariant Borel probability measure.

Theorem (Poincaré's recurrence theorem)

For μ -almost every $x \in X$,

$$\liminf_{n \rightarrow \infty} d(T^n x, x) = 0.$$

Remark: This is a **qualitative** result with no **quantitative** information.

Theorem (Boshernitzan, 1993)

Assume that for some $\alpha > 0$ the α -Hausdorff measure $\mathcal{H}^\alpha$ is σ -finite on (X, d) . Then for μ -almost every $x \in X$, we have

$$\liminf_{n \rightarrow \infty} n^{1/\alpha} d(T^n x, x) < \infty.$$

Moreover, if $\mathcal{H}^\alpha(X) = 0$, then for μ -almost every $x \in X$, we have

$$\liminf_{n \rightarrow \infty} n^{1/\alpha} d(T^n x, x) = 0.$$

Theorem (Barreira-Saussol, 2001)

If $T: X \rightarrow X$ is a Borel measurable map on $X \subset \mathbb{R}^d$, then for μ -almost every $x \in X$, we have

$$\liminf_{n \rightarrow \infty} n^{1/\alpha} d(T^n x, x) = 0 \quad \text{for any } \alpha > \liminf_{r \rightarrow 0} \frac{\log \mu(B(x, r))}{\log r}.$$

- Boshernitzan's result can be reformulated as: For μ -almost every x , there is a constant $c(x) > 0$ such that

$$d(T^n x, x) < c(x)n^{-1/\alpha} \quad \text{for i.m. } n \in \mathbb{N}.$$

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- How about the size of the set

$$\begin{aligned} \mathcal{R}(\psi) &= \{x \in X : d(T^n x, x) < \psi(n) \text{ for i.m. } n \in \mathbb{N}\} \\ &= \limsup_{n \rightarrow \infty} \{x \in X : d(T^n x, x) < \psi(n)\} \end{aligned}$$

defined by the asymptotic behavior of the orbits, where $\psi: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a non-negative function.

Known results

Size in measure:

- Chang-Wu-Wu (2019): Homogeneous self-similar sets.
- Baker-Farmer (2019): Self-conformal sets.
- Hussain-Li-Simmons-Wang (2022): Conformal dynamical system.
- Kirsebom-Kunde-Persson (2023): Integer matrix transformation (non-conformal).

Size in dimension:

- Tan-Wang (2011): Beta-dynamical system.
- Seuret-Wang (2015): Conformal iterated function systems.
- Yuan-Li (2023): Expanding Markov map.
- Hu-Persson (2023): “Cat map”.

Conformal case

Conditions:

- (1) (Ahlfors regularity) There exist $\eta \geq 1$ and $\delta > 0$ such that for any ball $B(x, r) \subset X$,

$$\eta^{-1}r^\delta \leq \mu(B(x, r)) \leq \eta r^\delta.$$

- (2) (Bounded distortion) There exists $K > 0$ such that

$$K^{-1} \leq \frac{d(T^n x, d^n y)}{d(x, y) \|D_x(T^n)\|} \leq K$$

for any $n \in \mathbb{N}$ and x, y in the same cylinder $J_n \in \mathcal{F}_n$.

- (3) (Conformality) There exists a constant $C > 0$ such that for any $J_n \in \mathcal{F}_n$ and ball $B(x, r) \subset J_n$,

$$B(T^n x, C^{-1} \|D_x(T^n)\| r) \subset T^n B(x, r) \subset B(T^n x, C \|D_x(T^n)\| r).$$

- (4) For all $n \in \mathbb{N}$, $\sum_{J_n \in \mathcal{F}_n} \left(\inf_{x \in J_n} \|D_x(T^n)\| \right)^{-\delta} \leq K_1$.

- (5) (**Exponential mixing 1**) There exist constants $c > 0$ and $\tau > 0$ such that for any **ball** $B \subset X$ and **measurable set** $E \subset X$,

$$|\mu(B \cap T^{-n}E) - \mu(B)\mu(E)| \leq ce^{-\tau n}\mu(E) \quad \text{for all } n \geq 1.$$

Theorem (Hussain-Li-Simmons-Wang, 2021)

Let ψ be a positive function on $\mathbb{N}$. Suppose that μ satisfies conditions (1)–(5). Then,

$$\mu(\mathcal{R}(\psi)) = \begin{cases} 0 & \text{if } \sum_{n \geq 1} \psi(n)^\delta < \infty, \\ 1 & \text{if } \sum_{n \geq 1} \psi(n)^\delta = \infty. \end{cases}$$

Remark: cannot directly apply to negative β -transformation.

Integer matrix transformation

Theorem (Kirsebom-Kunde-Persson, 2023)

Let $X = [0, 1]^d$, $T\mathbf{x} = A\mathbf{x} \pmod{1}$, where A is an **integer matrix** with the modulus of all eigenvalues strictly larger than 1. Then,

$$m_d(\mathcal{R}(\psi)) = \begin{cases} 0 & \text{if } \sum_{n \geq 1} \psi(n)^d < \infty, \\ 1 & \text{if } \sum_{n \geq 1} \psi(n)^d = \infty. \end{cases}$$

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- Does the zero-one law still hold if A is **not** an integer matrix?

Motivation

Fact:

$$d(T^n x, x) < \psi(n) \iff T^n x \in x + B(0, \psi(n)).$$

Question:

Balls replaced by hyperrectangles or hyperboloids?

Weighted Diophantine approximation

$$W_d(\Psi) = \{\mathbf{x} \in [0, 1]^d : |x_i - p_i/q| < \psi_i(q), 1 \leq i \leq d, \text{ for i.m. } (\mathbf{p}, q) \in \mathbb{Z}^d \times \mathbb{N}\},$$

where $\Psi = (\psi_1, \dots, \psi_d)$ and $\psi_i: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ for each $1 \leq i \leq d$.

$$|x_i - p_i/q| < \psi_i(q), 1 \leq i \leq d \iff \mathbf{x} \in \prod_{i=1}^d B(p_i/q, \psi_i(q)).$$

- How about the measure of

$$\mathcal{R}(\Psi) = \left\{ \mathbf{x} \in [0, 1]^d : T^n \mathbf{x} \in \mathbf{x} + \prod_{i=1}^d B(0, \psi_i(n)) \text{ for i.m. } n \in \mathbb{N} \right\}?$$

Multiplicative Diophantine approximation

$$W_d^\times(\delta_n) = \left\{ \mathbf{x} \in [0, 1]^d : \prod_{i=1}^d |x_i - p_i/q| < \delta_n \text{ for i.m. } (\mathbf{p}, q) \in \mathbb{Z}^d \times \mathbb{N} \right\},$$

where (δ_n) is a non-negative sequence.

$$\prod_{i=1}^d |x_i - p_i/q| < \delta_n \implies \mathbf{x} \in \mathbf{p}/q + H(\delta_n),$$

where $H(\delta_n) = \{\mathbf{z} \in [-1, 1]^d : |z_1 \cdots z_d| < \delta_n\}$.

- How about the measure of

$$\mathcal{R}^\times(\delta_n) = \{\mathbf{x} \in [0, 1]^d : T^n \mathbf{x} \in \mathbf{x} + H(\delta_n) \text{ for i.m. } n \in \mathbb{N}\}?$$

Shrinking target problems

Let (S_n) be a sequence subsets of X and let

$$W(T, \{S_n\}) := \{x \in X : T^n x \in S_n \text{ for i.m. } n \in \mathbb{N}\}.$$

- measure: Chernov-Kleinbock, Chazottes, Fayad, Galatolo-Rousseau-Saussol, Haydn-Nicol-Persson-Vaienti, Kim, Li-Liao-Velani-Zorin
- Hausdorff dimension: Hill-Velani, Urbański, Fernández-Melián-Pestana, Shen-Wang, Li-Wang-Wu-Xu, Koivusalo-Ramírez, Barany-Rams, Li-Liao-Velani-Zorin

Piecewise expanding maps

We say that $T: [0, 1]^d \rightarrow [0, 1]^d$ is a piecewise expanding map if there is a **finite** family $(U_i)_{i=1}^P$ of pairwise disjoint connected open subsets in $[0, 1]^d$ with $\bigcup_{i=1}^P \overline{U_i} = [0, 1]^d$ such that the following statements hold.

- (i) The map T is **injective** and can be extended to a **C^1 map** on each $\overline{U_i}$.
Moreover, there exists a constant **$L > 1$** such that

$$\inf_i \inf_{\mathbf{x} \in U_i} \|D_{\mathbf{x}}T\| \geq L,$$

where $D_{\mathbf{x}}T$ is the differential of T at $\mathbf{x}$ and

$$\|D_{\mathbf{x}}T\| := \sup_{\mathbf{z} \in \mathbb{R}^d} |(D_{\mathbf{x}}T)(\mathbf{z})|/|\mathbf{z}|.$$

- (ii) The boundary of U_i is included in a C^1 piecewise embedded compact submanifold of codimension one.

Exponential mixing

Definition (Exponential mixing)

For any collection $\mathcal{C}$ of measurable subsets of $[0, 1]^d$ satisfying the following **bounded property**

$$\sup_{F \in \mathcal{C}} M^{*(d-1)}(\partial F) < \infty,$$

there exist constants $c > 0$ and $\tau > 0$ such that for any $F, G \in \mathcal{C}$, we have

$$|\mu(F \cap T^{-n}G) - \mu(F)\mu(G)| \leq c\mu(G)e^{-\tau n}.$$

$M^{*(d-1)}(A)$ is the $(d-1)$ -dimensional upper Minkowski content defined by

$$M^{*(d-1)}(A) := \limsup_{\epsilon \rightarrow 0^+} \frac{m_d(\{\mathbf{x} \in \mathbb{R}^d : d(\mathbf{x}, A) < \epsilon\})}{\epsilon}.$$

Remark: If A is a $(d-1)$ -dimensional manifold, then $M^{*(d-1)}(A) = m_{d-1}(A)$.

Main results

Theorem 1 (H-Liao, 2023)

Let $T : [0, 1]^d \rightarrow [0, 1]^d$ be a piecewise expanding map. Suppose that μ is an absolutely continuous invariant measure satisfies the following conditions.

- (1) μ is exponential mixing.
- (2) The density h of μ belongs to $L^q(m_d)$ for some $q > 1$.

Then,

$$\mu(\mathcal{R}(\Psi)) = \begin{cases} 0 & \text{if } \sum_{n=1}^{\infty} \psi_1(n) \cdots \psi_d(n) < \infty, \\ 1 & \text{if } \sum_{n=1}^{\infty} \psi_1(n) \cdots \psi_d(n) = \infty. \end{cases}$$

- $m_d(\prod_{i=1}^d B(0, \psi_i(n))) \asymp \psi_1(n) \cdots \psi_d(n).$

Corollary 1

Let $T\mathbf{x} = A\mathbf{x} \pmod{1}$, where A is a **real** matrix with the modulus of all eigenvalues strictly larger than 1. Let μ be an absolutely continuous invariant measure. Suppose that A satisfies **one of** the following conditions.

- ❶ All eigenvalues of A are of modulus strictly larger than $1 + \sqrt{d}$.
- ❷ A is diagonal.
- ❸ A is an integer matrix.

Then,

$$\mu(\mathcal{R}(\Psi)) = \begin{cases} 0 & \text{if } \sum_{n=1}^{\infty} \psi_1(n) \cdots \psi_d(n) < \infty, \\ 1 & \text{if } \sum_{n=1}^{\infty} \psi_1(n) \cdots \psi_d(n) = \infty. \end{cases}$$

- holds for negative β -transformation.

Theorem 2 (H-Liao, 2023)

Let $T : [0, 1]^d \rightarrow [0, 1]^d$ be a piecewise expanding map. Suppose that μ is an absolutely continuous invariant measure satisfies the following conditions.

- (1) μ is exponential mixing.
- (2) There exists an open set V with $\mu(V) = 1$ such that the density h of μ , when restricted to V , is bounded from above by $c \geq 1$ and bounded from below by c^{-1} .

Then,

$$\mu(\mathcal{R}^\times(\delta_n)) = \begin{cases} 0 & \text{if } \sum_{n=1}^{\infty} \delta_n (-\log \delta_n)^{d-1} < \infty, \\ 1 & \text{if } \sum_{n=1}^{\infty} \delta_n (-\log \delta_n)^{d-1} = \infty. \end{cases}$$

- $m_d(H(\delta_n)) \asymp \delta_n (-\log \delta_n)^{d-1}$.

Corollary 2

Let $Tx = Ax \pmod{1}$, where A is a **real** matrix with the modulus of all eigenvalues strictly larger than 1. Let μ be an absolutely continuous invariant measure. Suppose that A satisfies **one of** the following conditions.

- ① A is diagonal with all eigenvalues belonging to $(-\infty, -(\sqrt{5} + 1)/2] \cup (1, +\infty)$.
- ② A is an integer matrix.

Then,

$$\mu(\mathcal{R}^\times(\delta_n)) = \begin{cases} 0 & \text{if } \sum_{n=1}^{\infty} \delta_n (-\log \delta_n)^{d-1} < \infty, \\ 1 & \text{if } \sum_{n=1}^{\infty} \delta_n (-\log \delta_n)^{d-1} = \infty. \end{cases}$$

Ideas and proofs

Setting:

- $d = 1$.
- $T : x \mapsto 2x \pmod{1}$.
- m_1 : Lebesgue measure.
- $\psi(n) \geq n^{-2}$ for all $n \in \mathbb{N}$.

Ideas and proofs

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- $T : x \mapsto 2x \pmod{1}$.
- m_1 : Lebesgue measure.
- $\psi(n) \geq n^{-2}$ for all $n \in \mathbb{N}$.

Chung-Erdős inequality: for measurable sets $A_1, \dots, A_n$,

$$m_1(A_1 \cup \dots \cup A_n) \geq \frac{(\sum_{j=1}^n \mu(A_j))^2}{\sum_{j,k=1}^n \mu(A_j \cap A_k)}.$$

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Goal: quasi-independence of the sets $E_n := \{x \in [0, 1] : |T^n x - x| < \psi(n)\}$.

Local structure of E_n

Lemma (Hussain-Li-Simmons-Wang, 2021)

Let $B = B(x, r)$ be a ball centered at $x \in [0, 1]$ and radius $r > 0$. Then for any $n \in \mathbb{N}$ with $\psi(n) > r$ and any subset F of B ,

$$F \cap T^{-n}B(x, \psi(n) - r) \subset F \cap E_n \subset F \cap T^{-n}B(x, \psi(n) + r).$$

Remark: This will lead to a superposition of the error terms, which is why we need $h \in L^q(m_1)$.

Upper estimates of $\mu(E_m \cap E_n)$

Partition $[0, 1]$ into n^2 intervals with equal length.

$$\begin{aligned}
 & \mu(E_m \cap E_n) \\
 & \leq \sum_{i \leq n^2} \mu(B(x_i, n^{-2}) \cap T^{-m} B(x_i, 2\psi(m)) \cap T^{-n} B(x_i, 2\psi(n))) \\
 & = \sum_{i \leq n^2} \mu(B(x_i, n^{-2}) \cap T^{-m} (B(x_i, 2\psi(m)) \cap T^{-(n-m)} B(x_i, 2\psi(n)))) \\
 & \leq \dots \\
 & \lesssim (1 + n^2 e^{-\tau m}) (\psi(m) + e^{-\tau(n-m)}) \cdot \psi(n).
 \end{aligned}$$

- $\psi(m) + n^{-2} \leq 2\psi(m)$ and $\psi(n) + n^{-2} \leq 2\psi(n)$.
- Works well for the case $m \geq (2 \log n)/\tau$.

$$m < (2 \log n) / \tau$$

Idea:

- apply the exponential mixing property to the sets $B(x_i, n^{-2}) \cap T^{-m}B(x_i, 2\psi(m))$ and $T^{-n}B(x_i, 2\psi(n))$.

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Obstacle:

- $B(x_i, n^{-2}) \cap T^{-m}B(x_i, 2\psi(m))$ may not satisfy bounded property.

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Simple observation:

- for any dyadic interval I_m of generation m , $B(x_i, n^{-2}) \cap T^{-m} B(x_i, 2\psi(m)) \cap I_m$ is either an empty set or an interval.

$$m < (2 \log n) / \tau$$

$$\begin{aligned}
 & \mu(B(x_i, n^{-2}) \cap T^{-m} B(x_i, 2\psi(m)) \cap T^{-n} B(x_i, 2\psi(n))) \\
 &= \sum_{I_m} \mu(B(x_i, n^{-2}) \cap T^{-m} B(x_i, 2\psi(m)) \cap I_m \cap T^{-n} B(x_i, 2\psi(n))) \\
 &\leq \sum_{I_m} \left(\mu(B(x_i, n^{-2}) \cap T^{-m} B(x_i, 2\psi(m)) \cap I_m) + ce^{-\tau n} \right) \mu(B(x_i, 2\psi(n))) \\
 &= \left(\mu(B(x_i, n^{-2}) \cap T^{-m} B(x_i, 2\psi(m))) + c2^m e^{-\tau n} \right) \mu(B(x_i, 2\psi(n))).
 \end{aligned}$$

Summing over $i \leq n^{-2}$, by a technical argument,

$$\begin{aligned}
 \mu(E_m \cap E_n) &\lesssim ((1 + cm^2 e^{-\tau m})\psi(m) + c2^m e^{-\tau n} n^2) \psi(n) \\
 &\lesssim \psi(m) \psi(n).
 \end{aligned}$$

Thank you!